Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy by dataset.

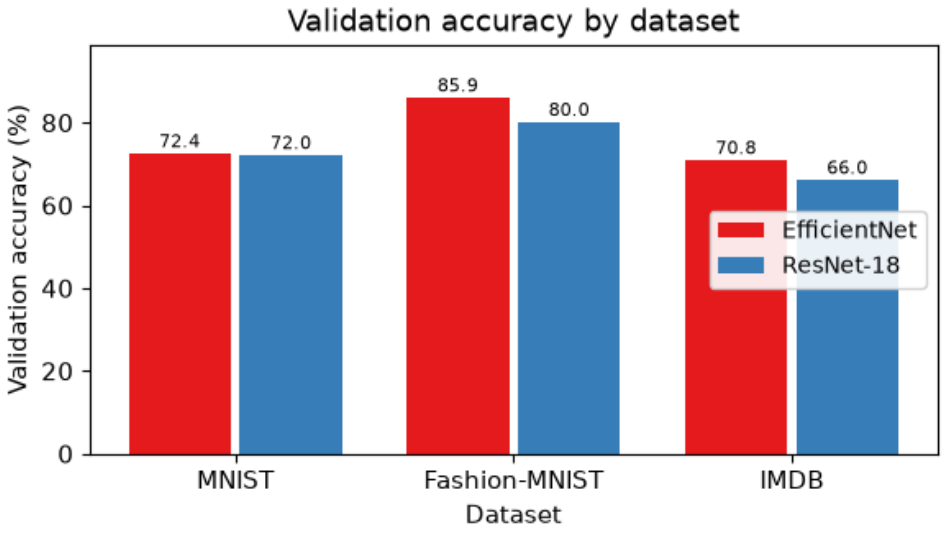


Figure 1: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 1, ResNet-18 achieves 66.0% validation accuracy on IMDB. On IMDB, EfficientNet obtains a validation accuracy of 70.8% (Figure 1).

Figure 2 shows the accuracy during training.

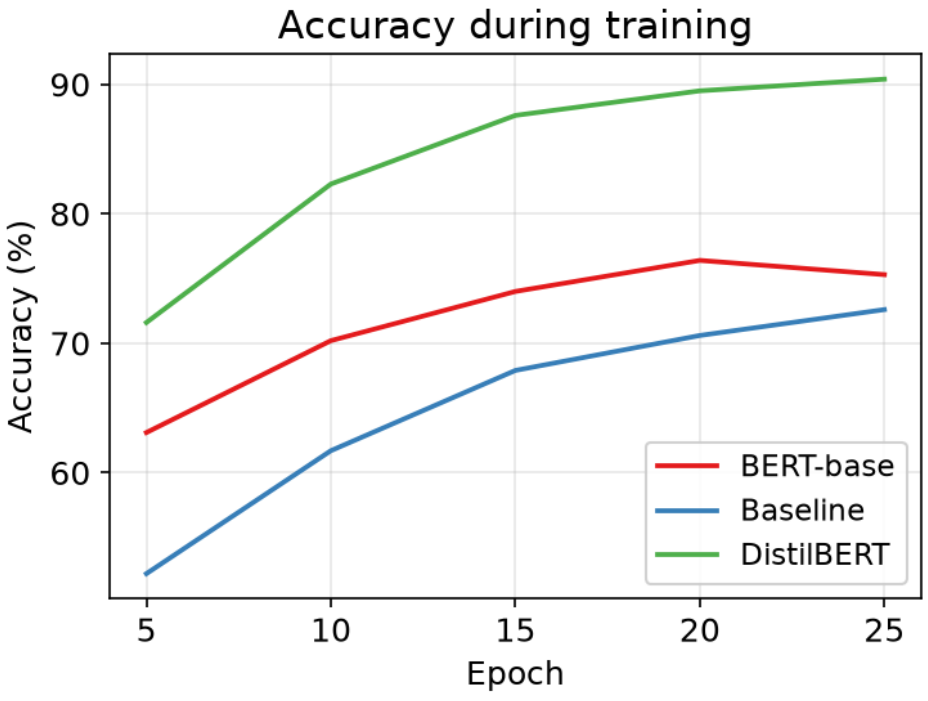


Figure 2: Accuracy during training for the models in this experiment.

As shown in Figure 2, at epoch 10, Baseline achieves 61.7% accuracy. At epoch 10, DistilBERT is 33.1 points better than BERT-base.

Figure 3 shows the validation accuracy during training.

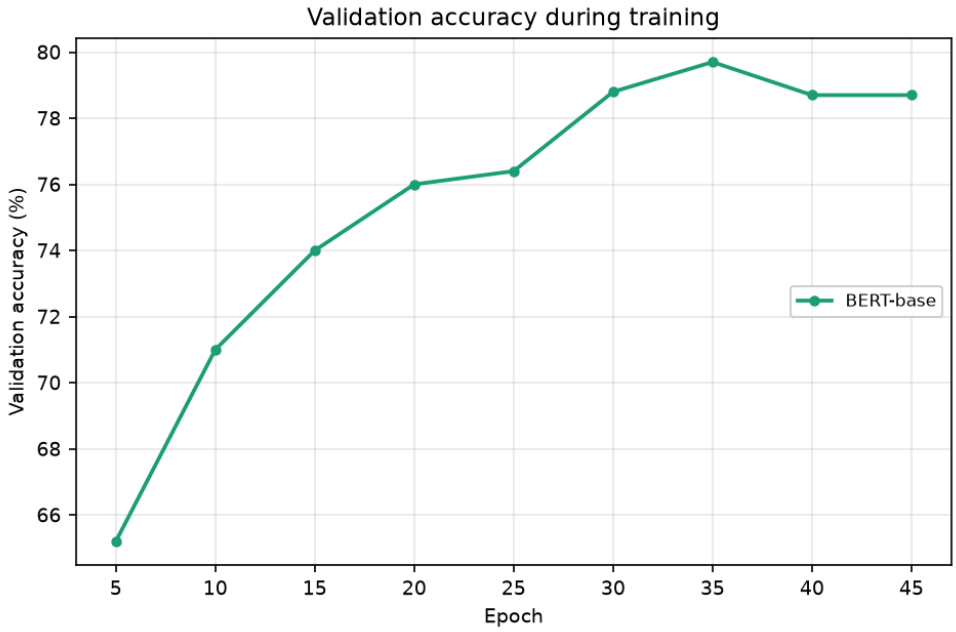


Figure 3: Validation accuracy during training for the models in this experiment.

BERT-base reaches 78.8% validation accuracy at epoch 30 (Figure 3). BERT-base scores 10.7 points higher on 40 than on 20 (Figure 3).

Figure 4 shows the f1 score by dataset.

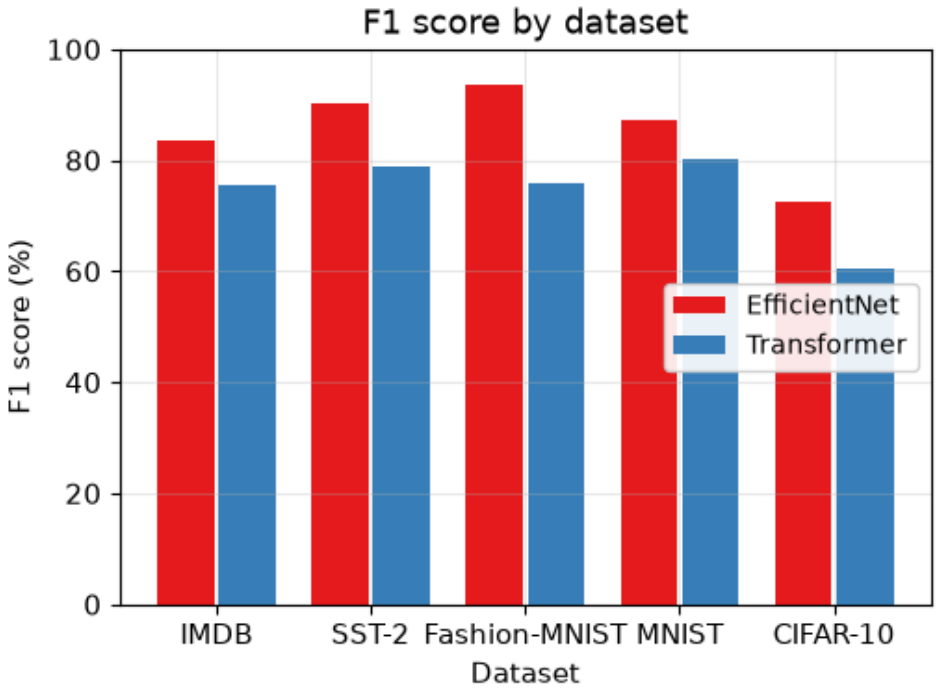


Figure 4: F1 score by dataset for the models in this experiment.

On MNIST, EfficientNet is 7 points better than Transformer (Figure 4). Transformer achieves 75.6% F1 score on IMDB.

3 Discussion

XGBoost reaches 65.3% accuracy on STL-10. The differences between the models are discussed in the next section.